OpenJLS

JPEG-LS Lossless Encoder IP Core — Datasheet

Version 1.0 • Single-component, fully pipelined, vendor-agnostic VHDL

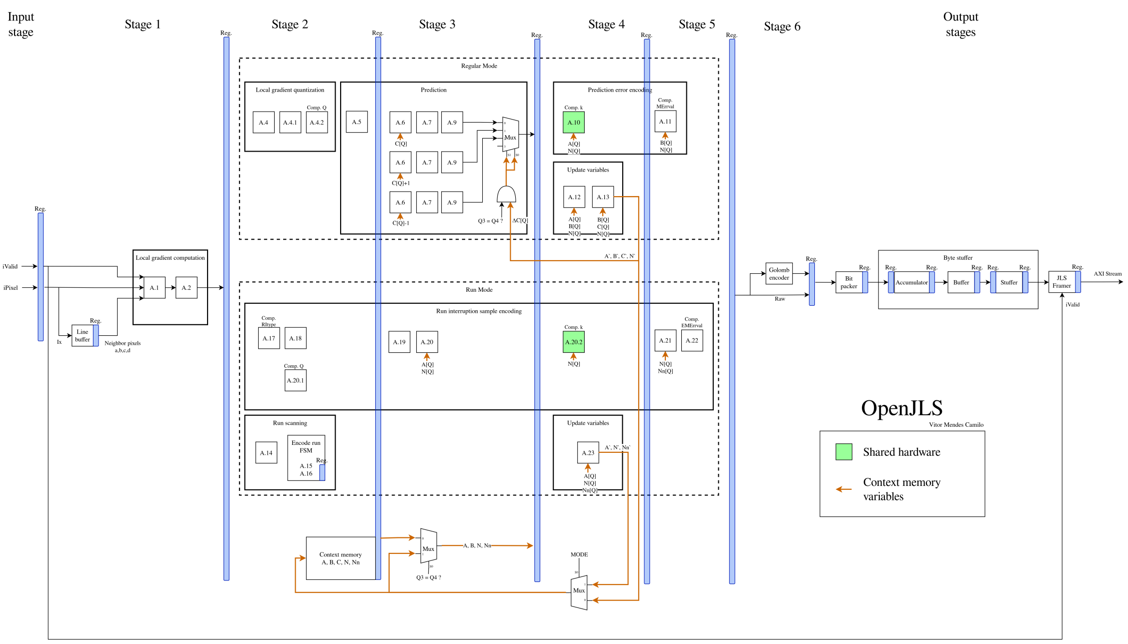
1 Overview

OpenJLS is a JPEG-LS (ISO/IEC 14495-1 / ITU-T T.87) lossless encoder for FPGAs. It encodes single-component (grayscale) images at one pixel per clock with no external memory, streaming a standards-compliant JPEG-LS bitstream out over AXI4-Stream.

Standard	JPEG-LS lossless (ISO/IEC 14495-1, ITU-T T.87)
Components	Single (grayscale)
Pixel depth	8–16 bits (`BITNESS`)
Image size	Up to 65535 × 65535 px, minimum 4 × 1
Throughput	1 pixel / clock
Latency	14 clock cycles
f_{max}	∼250 MHz (Xilinx UltraScale+ `xczu7eg-1`, RTL-only)
Interface	AXI4-Stream input (pixels) and output (bitstream)
Memory	On-chip line buffer, one image row; no external RAM
Resources	∼8k LUT, ∼2.1k FF + BRAM scaling with image width
Portability	Vendor-agnostic VHDL (open-logic primitives)

2 Functional description

The core is a fully pipelined encoder with a latency of 14 clock cycles: an input register, six processing stages (gradient computation, context modeling, MED prediction with bias correction, Golomb–Rice / run-length coding) and three internally registered output stages (bit packer, byte stuffer, framer). End-of-image is derived internally from the configured dimensions; the output bitstream is self-delimiting.



3 Generics

Generic	Range	Description
`BITNESS`	8–16	Pixel bit depth.
`MAX_IMAGE_WIDTH`	4–65535	Largest width supported; sets line-buffer depth.
`MAX_IMAGE_HEIGHT`	1–65535	Largest height supported.
`OUT_WIDTH`	48–1024	Output bus width in bits; multiple of 8.

4 Signals

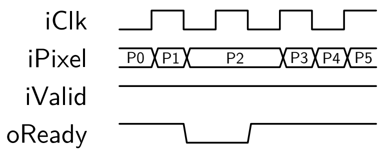
All ports are synchronous to the rising edge of `iClk`. The streaming ports map one-to-one onto AXI4-Stream as shown.

Port	Dir	Width	AXIS	Role
`iClk`	in	1	—	Clock.
`iRst`	in	1	—	Synchronous reset, active high; latches dimensions.
`iImageWidth`	in	$\lceil \log_2(\mathrm{MAX\_W}+1) \rceil$	—	Image width, pixels (configuration).
`iImageHeight`	in	$\lceil \log_2(\mathrm{MAX\_H}+1) \rceil$	—	Image height, pixels (configuration).
`iValid`	in	1	`TVALID`	Pixel valid.
`iPixel`	in	`BITNESS`	`TDATA`	Pixel data.
`oReady`	out	1	`TREADY`	Input ready.
`oData`	out	`OUT_WIDTH`	`TDATA`	Bitstream beat.
`oValid`	out	1	`TVALID`	Output valid.
`oKeep`	out	`OUT_WIDTH/8`	`TKEEP`	Valid-byte mask on final beat.
`oLast`	out	1	`TLAST`	End of image.
`iReady`	in	1	`TREADY`	Downstream ready.

The input omits `TLAST` (optional in AXI4-Stream); house-style port names map directly onto `s_axis`/`m_axis` for a thin wrapper.

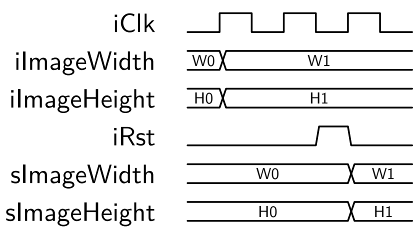
5 Timing and protocol

Pixel input. A pixel transfers on every cycle where `iValid` and `oReady` are both high. The core asserts `oReady` continuously and deasserts it only when the output FIFO fills under downstream backpressure.



`P2` is held flat across the stall and accepted when `oReady` returns high; no pixel is dropped while `oReady` is low.

Reset and configuration. The image dimensions are sampled while `iRst` is high: present `iImageWidth`/`iImageHeight` and hold them stable across a reset pulse, and the values latch into the internal `sImageWidth`/`sImageHeight` registers on the next clock. Pulse reset again before streaming a new resolution. Tying the dimension ports to `0` selects the `MAX_IMAGE_*` maxima; an out-of-range value falls back to the maximum with a simulation warning.



The new size (`W1/H1`) is driven for several cycles, then `iRst` pulses high; the dimensions register one clock after the reset is sampled.

Output. Beats appear on `oData/oValid` subject to `iReady`; `oLast` marks the final beat and `oKeep` flags its valid bytes. Pipeline latency is 14 clock cycles.

6 Resources and performance

Characterized on Xilinx Zynq UltraScale+ `xczu7eg-fbvb900-1-e` (speed grade −1), Vivado 2025.2, 12-bit, RTL-only (no floorplanning). f_{max} is true fmax (over-constrained until timing fails). At one pixel/clock, ∼250 MHz is ∼250 Mpixel/s.

f_{max} (MHz) vs image width

Width	Default	Perf.Exp.	Cong.Spread
4096	243.9	238.7	**258.0**
8192	240.0	243.0	**252.8**
12288	243.5	**248.5**	247.9
16384	235.9	244.1	**256.7**
32768	232.2	212.8	**247.5**

Resources vs image width

Width	LUT	FF	BRAM
4096	8048	2059	1.5
8192	7930	2062	3.0
12288	7855	2087	4.5
16384	7877	2089	5.5
32768	7990	2095	11.0

Logic is essentially constant with image size; only the line-buffer BRAM scales with width and pixel depth.

7 Integration example

```vhdl
u_enc : entity work.openjls_top
  generic map (
    BITNESS => 8,
    MAX_IMAGE_WIDTH => 4096,
    MAX_IMAGE_HEIGHT => 4096,
    OUT_WIDTH => 64 )
  port map (
    iClk => clk,
    iRst => rst, -- hold high while dims are stable
    iImageWidth => img_w, -- latched on reset
    iImageHeight => img_h,
    iValid => px_valid, -- pixel stream (AXIS slave)
    iPixel => px_data,
    oReady => px_ready,
    oData => m_tdata, -- bitstream (AXIS master)
    oValid => m_tvalid,
    oKeep => m_tkeep,
    oLast => m_tlast,
    iReady => m_tready );
```

8 Conformance and verification

Verified by simulation (NVC) with a layered suite: per-module OSVVM tests against T.87-derived reference models, top-level control-plane stress with requirements tracking, embedded PSL protocol contracts, 99%+ NVC statement coverage, and byte-exact comparison against the CharLS reference encoder over 287 images. The OSVVM stress test and a golden-model subset are re-run on the post-synthesis gate-level netlist. Live results: `https://vitormendesc.github.io/OpenJLS/`.

Revision history

Version	Date	Notes
1.0	2026-06	Initial release.

References. ITU-T T.87 / ISO/IEC 14495-1; Y. M. Mert, *Key Architectural Optimizations for Hardware Efficient JPEG-LS Encoder*, IEEE 2018; CharLS; open-logic.